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PISTON FOR DISC BRAKE DEVICE AND DISC BRAKE DEVICE

Abstract

A piston for a disc brake device includes: a cylindrical wall portion; a bottom portion; and a plurality of columnar ribs. The cylindrical wall portion includes an outer diameter side cylindrical portion including a seal sliding contact portion on an outer peripheral surface, and an inner diameter side cylindrical portion that includes a pad contact portion at an end portion on the other side in an axial direction. The plurality of columnar ribs are disposed to be inclined with respect to a piston central axis, and each have an end portion on one side in a length direction connected to a side surface on the other side in the axial direction of the bottom portion, and an end portion on the other side in the length direction connected to an inner peripheral surface of the inner diameter side cylindrical portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC §119 from Japanese Patent Application No. 2024-031028 filed on Mar. 1, 2024, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a piston for a disc brake device and a disc brake device.

BACKGROUND

[0003] Disc brake devices are widely used for braking automobiles and motorcycles. During braking performed by the disc brake device, a pair of pads disposed on both sides of a rotor rotating together with a wheel are pressed against both side surfaces of the rotor by a piston. Then, a kinetic energy of a vehicle is converted into a thermal energy by friction generated between the pads and the rotor, thereby braking the vehicle. Therefore, heat is accumulated in the rotor. The heat accumulated in the rotor is dissipated to the surroundings, and some of the heat is transferred through the pads to the piston, and then to a brake oil and a piston seal.

[0004] As such a disc brake device, a floating-type disc brake device, an opposed-piston-type disc brake device, and the like are known.

[0005] FIG. **10** illustrates an opposed-piston-type disc brake device **100** having a structure in the related art described in JP2011-179676A.

[0006] The disc brake device **100** includes a caliper **101** fixed to a vehicle body and a plurality of pistons **102**.

[0007] The caliper **101** includes an inner body **104** and an outer body **105** on both sides of a rotor **103** rotating together with a wheel. The inner body **104** and the outer body **105** each include a cylinder **106**.

[0008] The piston **102** is fitted to the cylinder **106**. The piston **102** and the cylinder **106** are sealed by a piston seal **107**.

[0009] The piston **102** has a bottomed cylindrical shape as a whole, and includes a cylindrical wall portion **108** having a substantially cylindrical shape and a bottom portion **109**. The bottom portion **109** closes an end portion on one side, which is disposed on a deep side of the cylinder **106**, of the cylindrical wall portion **108**.

[0010] In the structure in the related art described in JP2011-179676A, in order to make it difficult for heat transferred from pads **110***a* and **110***b* to the pistons **102** to be transferred to the piston seals **107**, a structure of the cylindrical wall portion **108** forming the piston **102** is devised.

[0011] Specifically, the cylindrical wall portion **108** includes an outer diameter side cylindrical portion **111** whose outer peripheral surface is in sliding contact with the piston seal **107**, an inner diameter side cylindrical portion **112** that presses the pads **110***a* and **110***b* with a tip end portion thereof, and an annular gap **113** formed between an inner peripheral surface of the outer diameter side cylindrical portion **111** and an outer peripheral surface of the inner diameter side cylindrical portion **112**.

[0012] During braking, the disc brake device **100** is supplied with the brake oil from a master cylinder to the cylinder **106**. Accordingly, the piston **102** fitted to the cylinder **106** of the inner body **104** is pushed out, and the pad **110***a* on an inner side supported by the caliper **101** is pressed against a side surface of the rotor **103**. Similarly, the piston **102** fitted to the cylinder **106** of the outer body **105** is pushed out, and the pad **110***b* on an outer side supported by the caliper **101** is pressed against a side surface of the rotor **103**. As a result, the rotor **103** is more strongly sandwiched by a pair of pads **110***a* and **110***b*, and the vehicle is braked.

[0013] In the disc brake device **100** having the structure in the related art, since the cylindrical wall portion **108** forming the piston **102** is divided into the outer diameter side cylindrical portion **111** that is in sliding contact with the piston seal **107** and the inner diameter side cylindrical portion **112** that presses the respective pads **110***a* and **110***b*, the heat is difficult to be transferred from the pads **110***a* and **110***b* to the piston seal **107**. Therefore, a temperature rise of the piston seal **107** can be prevented, and deterioration of the piston seal **107** can be prevented. Further, the heat of the pads **110***a* and **110***b* is difficult to be transferred to the brake oil, and a temperature rise of the brake oil can be prevented.

[0014] Patent Literature 1: JP2011-179676A

[0015] According to the disc brake device **100** having the structure in the related art described in JP2011-179676A, it is possible to prevent the temperature rise of the brake oil and to make it difficult for the heat to be transferred to the piston seal **107**. However, since the cylindrical wall portion **108** forming the piston **102** has a hollow structure, the rigidity may be insufficient.

[0016] On the one hand, since the disc brake device is provided on a road surface side with respect to a spring forming a suspension device in the vehicle, the disc brake device has a so- called unsprung load. Therefore, the piston forming the disc brake device is also required to be reduced in weight for a purpose of improving fuel efficiency and driving performance of the vehicle.

[0017] An object of the present disclosure is to provide a piston for a disc brake device that can prevent the temperature rise of the brake oil, make it difficult for heat to be transferred to a piston seal, prevent an increase in weight, and secure the rigidity.

SUMMARY

[0018] A piston for a disc brake device according to an aspect of the present disclosure includes: a cylindrical wall portion having a bottomed cylindrical shape as a whole and having a substantially cylindrical shape; a bottom portion configured to close an end portion on one side in an axial direction, which is configured to be disposed on a deep side of the cylinder, of the cylindrical wall portion; and a plurality of columnar ribs disposed on a radially inner side of the cylindrical wall portion.

[0019] The cylindrical wall portion includes an outer diameter side cylindrical portion including a seal sliding contact portion on an outer peripheral surface, an inner diameter side cylindrical portion that is disposed on a radially inner side of the outer diameter side cylindrical portion and that includes a pad contact portion at an end portion on the other side in the axial direction protruding in the axial direction with respect to an end portion on the other side in the axial direction of the outer diameter side cylindrical portion, and an annular gap formed between an inner peripheral surface of the outer diameter side cylindrical portion and an outer peripheral surface of the inner diameter side cylindrical portion.

[0020] The plurality of columnar ribs are disposed to be inclined with respect to a piston central axis, and each have an end portion on one side in a length direction connected to a side surface on the other side in the axial direction of the bottom portion, and an end portion on the other side in the length direction connected to an inner peripheral surface of the inner diameter side cylindrical portion.

[0021] In the piston for the disc brake device according an aspect of the present disclosure, each of the plurality of columnar ribs has the end portion on the other side in the length direction connected to an end portion on the other side in the axial direction of the inner peripheral surface of the inner diameter side cylindrical portion.

[0022] In the piston for the disc brake device according to an aspect of the present disclosure, the plurality of columnar ribs are disposed radially around the piston central axis.

[0023] In the piston for the disc brake device according to an aspect of the present disclosure, each of the plurality of columnar ribs has the end portion on the one side in the length direction

connected to an intermediate portion in the radial direction of the side surface on the other side in the axial direction of the bottom portion.
[0024] In the piston for the disc brake device according to an aspect of the present disclosure, the plurality of columnar ribs are disposed at equal intervals in a circumferential direction.
[0025] In the piston for the disc brake device according to an aspect of the present disclosure, the outer diameter side cylindrical portion, the inner diameter side cylindrical portion, and the plurality of columnar ribs are integrally formed.
[0026] In the piston for the disc brake device according to an aspect of the present disclosure, the outer diameter side cylindrical portion and the inner diameter side cylindrical portion are formed separately, and the inner diameter side cylindrical portion and the plurality of columnar ribs are integrally formed.
[0027] In the piston for the disc brake device according to an aspect of the present disclosure, the inner diameter side cylindrical portion has a communication hole penetrating in the radial direction.
[0028] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a portion, in which a phase in the circumferential direction coincides with that of the columnar rib, of the inner diameter side cylindrical portion.
[0029] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a vicinity of a connection portion, to which the end portion on the other side in the length direction of the columnar rib is connected, of the inner diameter side cylindrical portion.
[0030] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a portion adjacent to one side in the axial direction of the connection portion.
[0031] In the piston for the disc brake device according to an aspect of the present disclosure, at least a portion of the communication hole is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion.
[0032] In the piston for the disc brake device according to an aspect of the present disclosure, a central axis of the communication hole is inclined with respect to the piston central axis in a direction approaching the one side in the axial direction when extending toward the radially inner side.
[0033] In the piston for the disc brake device according to an aspect of the present disclosure, a heat shielding plate is provided on a radially inner side of the inner diameter side cylindrical portion, and the heat shielding plate covers the side surface on the other side in the axial direction of the bottom portion in a state of being in contact with the plurality of columnar ribs.
[0034] In the piston for the disc brake device according to an aspect of the present disclosure, the heat shielding plate is configured to be engaged with the plurality of columnar ribs.
[0035] In the piston for the disc brake device according to an aspect of the present disclosure, each of the plurality of columnar ribs has an engagement protrusion on a portion in the length direction thereof, the heat shielding plate has engagement holes, and the engagement protrusions are configured to be engaged with the respective engagement holes.
[0036] A disc brake device according to an aspect of the present disclosure includes: a caliper having a cylinder; and a piston fitted into the cylinder.
[0037] In the disk brake device according to an aspect of the present disclosure, the piston is the piston for the disc brake device according to an aspect of the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0038] FIG. **1** is a cross-sectional view illustrating a disc brake device according to a first embodiment of the present disclosure.

[0039] FIG. **2** is a front view illustrating a piston according to the first embodiment.

[0040] FIG. **3** is a cross-sectional view taken along a line A-A in FIG. **2**.

[0041] FIG. **4** is a perspective view illustrating the piston according to the first embodiment.

[0042] FIG. **5** is a front view illustrating a piston body according to the first embodiment.

[0043] FIG. **6** is a combination of a cross-sectional view taken along a line B-O-B in FIG. **5** and a cross-sectional view taken along a line C-O-C in FIG. **5**.

[0044] FIG. **7** is a perspective view illustrating the piston body according to the first embodiment.

[0045] FIG. **8** is a perspective view illustrating the piston body according to the first embodiment in a circumferential direction position different from that in FIG. **7**.

[0046] FIG. **9** is a cross-sectional view illustrating a piston body according to a second embodiment of the present disclosure.

[0047] FIG. **10** is a cross-sectional view illustrating a disc brake device having a structure in the related art.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0048] A first embodiment of the present disclosure will be described with reference to FIGS. **1** to **8**.

[0049] In the present embodiment, a case is described in which a piston for a disc brake device according to an aspect of the present disclosure is applied to an opposed-piston-type disc brake device used for braking an automobile.

[0050] However, the piston for a disc brake device according to an aspect of the present disclosure may be applied not only to the opposed-piston-type disc brake device, but also to other disc brake devices such as a floating-type disc brake device.

[0051] Hereinafter, an overall structure of a disc brake device **1** will be described, and then a structure of a piston **3** in the present embodiment will be described in detail.

[0052] In the following description related to the disc brake device **1**, unless otherwise specified, an axial direction, a circumferential direction, and a radial direction refer to an axial direction, a circumferential direction, and a radial direction of a disc-shaped rotor **5** (see FIG. **1**) rotating together with a wheel. In FIG. **1**, a left-right direction corresponds to the axial direction, a front-back direction corresponds to the circumferential direction, and an upper-lower direction corresponds to the radial direction. Further, a right side in FIG. **1**, which is a center side of a vehicle body in an assembled state to the vehicle body, is referred to as an axially inner side, and a left side in FIG. **1**, which is an outer side of the vehicle body in the assembled state to the vehicle body, is referred to as an axially outer side.

[0053] The disc brake device **1** in the present embodiment includes a caliper **2**, a plurality of pistons **3**, and a pair of pads **4***a* and **4***b*.

[0054] The caliper **2** is supported and fixed to a knuckle of a suspension device in a state in which the caliper **2** covers the rotor **5** from the radially outer side. The caliper **2** supports the pair of pads **4***a* and **4***b* to be movable in the axial direction.

[0055] The caliper **2** is integrally formed by subjecting a material made of a light alloy such as an aluminum alloy or an iron-based alloy to a casting process or the like.

[0056] The caliper **2** includes an inner body **6** and an outer body **7** disposed to sandwich the rotor **5**, and a bridge **8** connecting the inner body **6** and the outer body **7** in the axial direction.

[0057] The inner body **6** is disposed on the center side of the vehicle body with respect to the rotor **5** in the axial direction. The outer body **7** is disposed on the outer side of the vehicle body with respect to the rotor **5** in the axial direction. The bridge **8** is disposed on the radially outer side of the rotor **5**.

[0058] The inner body **6** and the outer body **7** each include a cylinder **9**. The cylinder **9** has a substantially cylindrical space. The cylinder **9** provided in the inner body **6** is open to an axially outer surface of the inner body **6**. The cylinder **9** provided in the outer body **7** is open to an axially inner surface of the outer body **7**. The inner body **6** and the outer body **7** each have an oil passage (not illustrated) for supplying a brake oil from a master cylinder to the cylinder **9**. The inner body **6** includes a connection portion **10** for connecting a brake hose between the inner body **6** and the master cylinder.

[0059] The inner body **6** and the outer body **7** each have, around a portion on an opening side of the cylinder **9**, a seal groove **11** having an annular shape. A piston seal **12** made of an elastic material is attached to the seal groove **11**.

[0060] The piston **3** is fitted to the cylinder **9**. Specifically, the pistons **3** are fitted to the respective cylinders **9** provided in the inner body **6** and the outer body **7**. The piston **3** and the cylinder **9** are sealed by the piston seal **12**.

[0061] In the present embodiment, a case is illustrated in which the piston for a disc brake device according to an aspect of the present disclosure is fitted to each of a cylinder provided in an inner body and a cylinder provided in an outer body. However, the piston for an opposed-type disc brake device according to an aspect of the present disclosure may be fitted to only one cylinder of the cylinder provided in the inner body and the cylinder provided in the outer body.

[0062] The pair of pads **4***a* and **4***b* each include a lining **13** and a metal back plate **14** supporting a back surface of the lining **13**. The back surface of the lining **13** refers to an axially inner surface for the lining **13** of the pad **4***a* on an inner side, and refers to an axially outer surface for the lining **13** of the pad **4***b* on an outer side. The pad **4***a* on the inner side of the pair of pads **4***a* and **4***b*, which is disposed on the axially inner side of the rotor **5**, is supported by the inner body **6** to be movable in the axial direction. The pad **4***b* on the outer side of the pair of pads **4***a* and **4***b*, which is disposed on the axially outer side of the rotor **5**, is supported by the outer body **7** to be movable in the axial direction.

[0063] During braking, the disc brake device **1** is supplied with the brake oil from the master cylinder to the cylinder **9**. Accordingly, the piston **3** fitted to the cylinder **9** of the inner body **6** is pushed out, and the lining **13** of the pad **4***a* on the inner side is pressed against an axially inner surface of the rotor **5**. Further, since the cylinder **9** of the inner body **6** and the cylinder **9** of the outer body **7** are connected by a tubing or the like (not illustrated) and form a hydraulic circuit, similarly, the piston **3** fitted to the cylinder **9** of the outer body **7** is pushed out, and the lining **13** of the pad **4***b* on the outer side is pressed against an axially outer surface of the rotor **5**. As a result, the rotor **5** is strongly sandwiched by the linings **13** of the pair of pads **4***a* and **4***b* in the axial direction, and a vehicle is braked.

[0064] The structure of the piston **3** in the present embodiment will be described in detail below.

[0065] In the following description, unless otherwise specified, the axial direction, the circumferential direction, and the radial direction refer to an axial direction, a circumferential direction, and a radial direction of the piston **3**. The axial direction of the piston **3** coincides with the axial direction of the rotor **5**. In a state in which the piston **3** is fitted to the cylinder **9**, a deep side of the cylinder **9** is referred to as one side in the axial direction, and an opening side of the cylinder **9** is referred to as the other side in the axial direction. That is, in the piston **3** fitted to the cylinder **9** of the inner body **6**, the axially inner side corresponds to the one side in the axial direction, and the axially outer side corresponds to the other side in the axial direction. In contrast, in the piston **3** fitted to the cylinder **9** of the outer body **7**, the axially outer side corresponds to the one side in the axial direction, and the axially inner side corresponds to the other side in the axial direction.

[Structure of Piston]

[0066] The piston **3** has a piston central axis O.sub.3 and has a bottomed cylindrical shape as a

whole.
[0067] The piston **3** includes a cylindrical wall portion **15**, a bottom portion **16**, and a plurality of columnar ribs **17**.
[0068] The piston **3** is manufactured by a selective laser sintering (SLS) method using a metal powder of a titanium alloy, an aluminum alloy, an iron alloy, or the like. However, the piston for a disc brake device according to an aspect of the present disclosure may be manufactured by using various three-dimensional object shaping methods {for example, a fused deposition modeling method, an inkjet method, a powder binding method, a stereolithography method, a laser engineered net shaping (LENS) method, a fused deposition modeling (FDM) method}.
[0069] In the piston **3** in the present embodiment, the cylindrical wall portion **15**, the bottom portion **16**, and the plurality of columnar ribs **17** are integrally formed. The cylindrical wall portion **15**, the bottom portion **16**, and the plurality of columnar ribs **17** form a piston body **18** which is one component.
[0070] In a non-braking state, an end portion on the other side in the axial direction of cylindrical wall portion **15** of the piston **3** is exposed from the cylinder **9**.
[0071] The cylindrical wall portion **15** has a substantially cylindrical shape and is hollow. In the present embodiment, the cylindrical wall portion **15** has a substantially U-shaped cross section.
[0072] The cylindrical wall portion **15** includes an outer diameter side cylindrical portion **19**, an inner diameter side cylindrical portion **20**, and an annular gap **21**.
[0073] In the present embodiment, since the cylindrical wall portion **15** and the plurality of columnar ribs **17** are integrally formed, the outer diameter side cylindrical portion **19**, the inner diameter side cylindrical portion **20**, and the plurality of columnar ribs **17** are integrally formed.
[0074] The outer diameter side cylindrical portion **19** forms a radially outer side portion of the cylindrical wall portion **15** and has a substantially cylindrical shape. An outer peripheral surface of the outer diameter side cylindrical portion **19** forms an outer peripheral surface of the cylindrical wall portion **15** and has a cylindrical surface shape. An outer diameter of the outer diameter side cylindrical portion **19** is slightly smaller than an inner diameter of the cylinder **9**. A radial thickness of the outer diameter side cylindrical portion **19** is substantially constant in the axial direction.
[0075] The outer diameter side cylindrical portion **19** includes a seal sliding contact portion **22** on the outer peripheral surface. The seal sliding contact portion **22** is provided at an intermediate portion in the axial direction of the outer peripheral surface of the outer diameter side cylindrical portion **19**. The seal sliding contact portion **22** is in sliding contact with the piston seal **12** in the axial direction.
[0076] The outer diameter side cylindrical portion **19** has, at an end portion on the other side in the axial direction of an inner peripheral surface, a tapered surface **23** whose inner diameter increases toward the other side in the axial direction.
[0077] The inner diameter side cylindrical portion **20** forms a radially inner side portion of the cylindrical wall portion **15** and has a substantially cylindrical shape. The inner diameter side cylindrical portion **20** is located on a radially inner side of the outer diameter side cylindrical portion **19** and is disposed such that most of the inner diameter side cylinder portion **20** overlaps with the outer diameter side cylindrical portion **19** in the radial direction. The inner diameter side cylindrical portion **20** is disposed coaxially with the outer diameter side cylindrical portion **19**. The inner diameter side cylindrical portion **20** and the outer diameter side cylindrical portion **19** are separated from each other in the radial direction, and are connected only at end portions on the one side in the axial direction.
[0078] In the present embodiment, an outer peripheral surface of the inner diameter side cylindrical portion **20** is a concave curved surface having a concave arc-shaped cross section, in which an outer diameter smoothly decreases toward an intermediate portion in the axial direction. Most of an inner peripheral surface of the inner diameter side cylindrical portion **20** is a convex curved surface having a convex arc-shaped cross section, in which an inner diameter smoothly decreases toward

the intermediate portion in the axial direction. However, when the piston for a disc brake device according to an aspect of the present disclosure is implemented, the outer peripheral surface and the inner peripheral surface of the inner diameter side cylindrical portion may have the cylindrical surface shape.
[0079] In the present embodiment, the end portion on the one side in the axial direction of the inner diameter side cylindrical portion **20** is offset towards the other side in the axial direction with respect to the end portion on the one side in the axial direction of the outer diameter side cylindrical portion **19**. In other words, the end portion on the one side in the axial direction of the outer diameter side cylindrical portion **19** protrudes in the axial direction with respect to the end portion on the one side in the axial direction of the inner diameter side cylindrical portion **20**. Specifically, the end portion on the one side in the axial direction of the inner diameter side cylindrical portion **20** is connected to a portion, which is closer to one side in the axial direction, of the inner peripheral surface of the outer diameter side cylindrical portion **19**.
[0080] A portion on the other side in the axial direction of the inner diameter side cylindrical portion **20** protrudes in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19**. The inner diameter side cylindrical portion **20** includes a pad contact portion **24** at an end portion on the other side in the axial direction. In the present embodiment, in the non-braking state, a portion including the pad contact portion **24**, which protrudes in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19**, of the inner diameter side cylindrical portion **20** is exposed from the cylinder **9**.
[0081] In the present embodiment, since the pad contact portion **24** protrudes in the axial direction from the outer diameter side cylindrical portion **19**, the outer diameter side cylindrical portion **19** does not directly contact the pads **4***a* and **4***b*. Therefore, it is possible to effectively prevent heat of the pads **4***a* and **4***b* from being directly transferred to the outer diameter side cylindrical portion **19** and reaching the brake oil. A portion, to which the heat is directly transferred due to contact with the pads **4***a* and **4***b*, of the piston **3** is a pad contact portion **24**. Therefore, in order to transfer the heat from the pad contact portion **24** to the piston seal **12**, the heat needs to be transferred through the inner diameter side cylindrical portion **20** and the columnar ribs **17** to the outer diameter side cylindrical portion **19** via the bottom portion **16**, and a heat transfer distance is sufficiently long. Therefore, the heat transferred to the piston seal **12** is sufficiently prevented.
[0082] The pad contact portions **24** are in contact with the respective back plates **14** of the pads **4***a* and **4***b*. A radial thickness of the pad contact portion **24** is larger than a radial thickness of a portion, which is separated from the pad contact portion **24**, of the inner diameter side cylindrical portion **20**. The radial thickness of the portion, which is separated from the pad contact portion **24**, of the inner diameter side cylindrical portion **20** is substantially the same as the radial thickness of the outer diameter side cylindrical portion **19**.
[0083] An end surface on the other side in the axial direction of the pad contact portion **24** has a circular ring shape and is a flat surface on a virtual plane orthogonal to the piston central axis O.sub.3.
[0084] The annular gap **21** is an annular gap portion formed between the inner peripheral surface of the outer diameter side cylindrical portion **19** and the outer peripheral surface of the inner diameter side cylindrical portion **20**, and forms a so-called heat insulation and heat dissipation portion. That is, a heat insulation effect can be expected by an air layer present in the annular gap **21**, and heat dissipation is performed by movement of the air layer. The annular gap **21** is open to the other side in the axial direction. An end portion on the one side in the axial direction of the annular gap **21** (a bottom portion in the axial direction) is located on the one side in the axial direction with respect to the seal sliding contact portion **22**.
[0085] A radial thickness of the annular gap **21** is substantially constant in the axial direction, but a radial thickness of a portion in which the tapered surface **23** is provided on the inner peripheral

surface of the outer diameter side cylindrical portion **19** is larger than a radial thickness of the other portion.
[0086] The bottom portion **16** closes an end portion on the one side in the axial direction, which is disposed on the deep side of the cylinder **9**, of the cylindrical wall portion **15**. The bottom portion **16** has a substantially disc-shape. An outer peripheral edge portion of the bottom portion **16** is connected to the end portion on the other side in the axial direction of the inner diameter side cylindrical portion **20** forming the cylindrical wall portion **15**.
[0087] A side surface on the one side in the axial direction of the bottom portion **16** is a partially spherical convex surface slightly bulging to the one side in the axial direction. A side surface on the other side in the axial direction of the bottom portion **16** is a partially spherical concave surface slightly recessed toward the one side in the axial direction. However, when the piston for a disc brake device according to an aspect of the present disclosure is implemented, a bottom portion may have a flat plate shape.
[0088] An axial thickness of the bottom portion **16** is substantially constant in the radial direction and the circumferential direction. In the present embodiment, the axial thickness of the bottom portion **16** is substantially the same as the radial thickness of the portion, which is separated from the pad contact portion **24**, of the inner diameter side cylindrical portion **20** and the radial thickness of the outer diameter side cylindrical portion **19**.
[0089] The plurality of columnar ribs **17** have a function of ensuring the rigidity of the piston **3**. Specifically, the plurality of columnar ribs **17** improve the rigidity of the inner diameter side cylindrical portion **20**, which presses the respective pads **4***a* and **4***b* during braking, of the cylindrical wall portion **15** forming the piston **3**.
[0090] Although the number of the columnar ribs **17** is not limited thereto, the piston **3** may include, for example, **3** to **12** columnar ribs **17**. In the present embodiment, the piston **3** includes six columnar ribs **17**.
[0091] The plurality of columnar ribs **17** are disposed on a radially inner side of the cylindrical wall portion **15**. Specifically, the plurality of columnar ribs **17** are disposed in a piston internal space **25** that is present on a radially inner side of the inner diameter side cylindrical portion **20** forming the cylindrical wall portion **15**.
[0092] The columnar rib **17** has a straight elongated shape. In the present embodiment, the columnar rib **17** has a substantially trapezoidal cross section and has a quadrangular prism shape whose cross-sectional shape changes in a length direction. However, when the piston for a disc brake device according to an aspect of the present disclosure is implemented, the columnar rib is not particularly limited in a shape thereof, and may have any shape such as a triangular column shape, a rectangular column shape, a trapezoidal column shape, a polygonal column shape such as a pentagonal column or a hexagonal column, a columnar shape, an elliptical column shape, a truncated conical shape, or a polygonal pyramid shape.
[0093] The columnar rib **17** is solid as a whole. However, a columnar rib may be hollow.
[0094] The plurality of columnar ribs **17** are disposed to be inclined with respect to the piston central axis O.sub.3. That is, a central axis O.sub.17 (see FIG. **3**) of the columnar rib **17** is inclined with respect to the piston central axis O.sub.3.
[0095] The plurality of columnar ribs **17** each have an end portion on one side in the length direction connected to the side surface on the other side in the axial direction of the bottom portion **16**, and an end portion on the other side in the length direction connected to the inner peripheral surface of the inner diameter side cylindrical portion **20**. Therefore, the central axis O.sub.17 of the columnar rib **17** is inclined with respect to the piston central axis O.sub.3 in a direction approaching the radially outer side from the one side in the length direction toward the other side in the length direction.
[0096] In the present embodiment, the plurality of columnar ribs **17** each have the end portion on the one side in the length direction connected to an intermediate portion in the radial direction of

the side surface on the other side in the axial direction of the bottom portion **16**, and the end portion on the other side in the length direction connected to an end portion on the other side in the axial direction of the inner peripheral surface of the inner diameter side cylindrical portion **20**. That is, the end portion on the other side in the length direction of the columnar rib **17** is connected to the pad contact portion **24**.
[0097] The columnar rib **17** is not connected to the bottom portion **16** and the inner diameter side cylindrical portion **20** at a portion other than the end portions on both sides in the length direction.
[0098] In the present embodiment, the plurality of columnar ribs **17** are disposed at equal intervals in the circumferential direction, and are disposed radially around the piston central axis O.sub.3. The plurality of columnar ribs **17** are separated from each other in the circumferential direction, and are independent of each other. In other words, two columnar ribs **17** adjacent to each other in the circumferential direction are not connected to each other. However, two columnar ribs adjacent to each other in the circumferential direction may be connected to each other in the circumferential direction.
[0099] In the present embodiment, a thickness t (see FIG. **3**) of the columnar rib **17** in the radial direction decreases from the one side in the length direction toward the other side in the length direction.
[0100] A width w (see FIG. **5**) of the columnar rib **17** in the circumferential direction increases from the one side in the length direction toward the other side in the length direction, except for the end portions on both sides in the length direction of the columnar rib **17**.
[0101] A first reinforcing portion **26***a* protruding in the radial direction and the circumferential direction is provided at the end portion on the one side in the length direction of the columnar rib **17**. A second reinforcing portion **26***b* protruding in the circumferential direction is provided at the end portion on the other side in the length direction of the columnar rib **17**.
[0102] In the present embodiment, an engagement protrusion **27** for attaching a heat shielding plate **29** described later is formed on a portion of the columnar rib **17** in the length direction. The engagement protrusion **27** is formed on a radially inner surface of an intermediate portion in the length direction of the columnar rib **17**, and protrudes toward the radially inner side. In the present embodiment, the engagement protrusion **27** is formed on all the columnar ribs **17**, respectively, but it is not necessary that the engagement protrusion **27** is formed on all the columnar ribs **17**, respectively.
[0103] The engagement protrusion **27** has a substantially triangular plate shape. However, the engagement protrusion is not limited to a triangular plate shape, and may have other shape such as a hemispherical shape.
[0104] In the piston **3** in the present embodiment, the inner diameter side cylindrical portion **20** has communication holes **28** penetrating in the radial direction. The communication hole **28** allows the annular gap **21** to communicate with the piston internal space **25**. However, when the piston for a disc brake device according to an aspect of the present disclosure is implemented, it is optional to form a communication hole in the inner diameter side cylindrical portion.
[0105] The inner diameter side cylindrical portion **20** has a plurality of communication holes **28**. In the present embodiment, the inner diameter side cylindrical portion **20** has the same number of the communication holes **28** as the columnar ribs **17**. However, the number of the communication holes formed in the inner diameter side cylindrical portion may not be the same as the number of the columnar ribs.
[0106] In the present embodiment, each of the plurality of communication holes **28** is formed in a portion, in which a phase in the circumferential direction coincides with that of the respective columnar rib **17**, of the inner diameter side cylindrical portion **20**. Therefore, the plurality of communication holes **28** are formed at equal intervals in the circumferential direction.
[0107] The communication hole **28** is formed in a vicinity of a connection portion, to which the end portion on the other side in the length direction of the columnar rib **17** is connected, of the inner

diameter side cylindrical portion **20**. In the present embodiment, the communication hole **28** is formed in a portion, which is adjacent to one side in the axial direction of the connection portion to which the end portion on the other side in the length direction of the columnar rib **17** is connected, of the inner diameter side cylindrical portion **20**. Here, since the end portion on the other side in the length direction of the columnar rib **17** is connected to the end portion on the other side in the axial direction of the inner peripheral surface of the inner diameter side cylindrical portion **20**, the communication hole **28** is formed in a portion that is closer to the other side in the axial direction of the inner diameter side cylindrical portion **20**.
[0108] At least a portion of the communication hole **28** is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19**. In the present embodiment, a half portion of the communication hole **28** on the other side in the axial direction is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19**. That is, the half portion of the communication hole **28** on the other side in the axial direction is a portion exposed from the outer diameter side cylindrical portion **19**. Therefore, in the non-braking state, the half portion of the communication hole **28** on the other side in the axial direction is exposed from the cylinder **9**. The half portion of the communication hole **28** on the one side in the axial direction is a portion covered by the outer diameter side cylindrical portion **19**. Specifically, the half portion of the communication hole **28** on the one side in the axial direction is covered by the outer diameter side cylindrical portion **19** from the radially outer side.
[0109] In the present embodiment, the communication hole **28** has a substantially rectangular opening shape as viewed in the radial direction. However, the opening shape of the communication hole is not limited to a rectangular shape, and may have any shape such as a circle, an ellipse, a triangle, or a polygon.
[0110] A central axis O.sub.28 (see FIG. **6**) of the communication hole **28** is inclined with respect to the piston central axis O.sub.3 in a direction approaching the one side in the axial direction when extending toward the radially inner side. Therefore, an extension line of the central axis O.sub.28 of the communication hole **28** passes through the side surface on the other side in the axial direction of the bottom portion **16**.
[0111] The piston **3** in the present embodiment further includes a heat shielding plate **29** in addition to the piston body **18**. The heat shielding plate **29** has a function of preventing the heat from being transferred from the pads **4***a* and **4***b* to the bottom portion **16**. However, when the piston for a disc brake device according to an aspect of the present disclosure is implemented, it is optional to provide a heat shielding plate.
[0112] The heat shielding plate **29** is provided separately from the cylindrical wall portion **15**, the bottom portion **16**, and the columnar ribs **17**. That is, the heat shielding plate **29** is provided separately from the piston body **18**.
[0113] The heat shielding plate **29** is made of a metal plate. In the present embodiment, the heat shielding plate **29** is manufactured by pressing a stainless steel plate.
[0114] The heat shielding plate **29** is disposed on the radially inner side of the inner diameter side cylindrical portion **20** forming the cylindrical wall portion **15**. That is, the heat shielding plate **29** is disposed in the piston internal space **25**.
[0115] The heat shielding plate **29** covers the side surface on the other side in the axial direction of the bottom portion **16** in a state of being in contact with the plurality of columnar ribs **17**. In the present embodiment, the heat shielding plate **29** is in contact with the plurality of columnar ribs **17** by engaging with the plurality of columnar ribs **17**.
[0116] The heat shielding plate **29** includes a lid portion **30** having a substantially disc-shape and a plurality of engagement pieces **31**.
[0117] In the present embodiment, the number of the engagement pieces **31** of the heat shielding

plate **29** is half that of the columnar ribs **17**. However, the number of engagement pieces provided on the heat shielding plate is not limited to half that of the columnar ribs. The plurality of engagement pieces **31** are disposed at equal intervals in the circumferential direction.

[0118] The engagement piece **31** has a substantially rectangular plate shape, and a base end portion of the engagement piece **31** is connected to an outer peripheral edge portion of the lid portion **30**. The engagement piece **31** is bent in a direction approaching the piston central axis O.sub.3 from a base end side toward a tip end side.

[0119] The engagement piece **31** has an engagement hole **32** having a hole shape, which can be engaged with the engagement protrusion **27**. In the present embodiment, the engagement hole **32** is a rectangular hole.

[0120] The heat shielding plate **29** cannot fall off into the piston internal space **25** by engaging the engagement protrusions **27** with the respective engagement holes **32** formed in the plurality of engagement pieces **31**.

[0121] In the present embodiment, the lid portion **30** has a notch **33**, into which a portion on the other side in the axial direction of the columnar rib **17** can be inserted, between two engagement pieces **31** adjacent to each other in the circumferential direction in the outer peripheral edge portion. The lid portion **30** has, at the center thereof, a small hole **34** into which a tip end portion of a tool can be engaged when the heat shielding plate **29** is removed.

[0122] According to the piston **3** in the present embodiment, it is possible to prevent a temperature rise of the brake oil, to make it difficult for the heat to be transferred to the piston seal **12**, to prevent an increase in weight, and to secure the rigidity.

[0123] That is, the cylindrical wall portion **15** forming the piston **3** in the present embodiment is divided into the outer diameter side cylindrical portion **19** including the seal sliding contact portion **22** on the outer peripheral surface, and the inner diameter side cylindrical portion **20** including the pad contact portion **24** at the end portion on the other side in the axial direction. Therefore, the heat of the pads **4***a* and **4***b* is difficult to be transferred to the seal sliding contact portion **22**. That is, in the piston **3** in the present embodiment, since a heat transfer distance from the pad contact portion **24** to the seal sliding contact portion **22** is longer than that in a case in which a seal sliding contact portion and a pad contact portion are provided in a same cylindrical portion, the heat of the pads **4***a* and **4***b* is difficult to be transferred to the seal sliding contact portion **22**. In particular, in the present embodiment, since the inner diameter side cylindrical portion **20** and the outer diameter side cylindrical portion **19** are connected only at the end portions on the one side in the axial direction, the heat transfer distance is sufficiently long, and the heat of the pads **4***a* and **4***b* is difficult to be transferred to the seal sliding contact portion **22**. Therefore, it is possible to make it difficult for the heat to be transferred to the piston seal **12**. As a result, a temperature rise of the piston seal **12** is prevented, and the deterioration of the piston seal **12** is prevented.

[0124] Since the piston **3** in the present embodiment includes the plurality of columnar ribs **17** in order to ensure the rigidity, it is possible to prevent the increase in weight and to secure the rigidity of the piston **3**. In particular, the plurality of columnar ribs **17** are each disposed to be inclined with respect to the piston central axis O.sub.3, and have the end portion on the one side in the length direction connected to the side surface on the other side in the axial direction of the bottom portion **16**, and the end portion on the other side in the length direction connected to the inner peripheral surface of the inner diameter side cylindrical portion **20**. Therefore, since the plurality of columnar ribs **17** are stretched between the inner diameter side cylindrical portion **20** that includes the pad contact portion **24** pressing the respective pads **4***a* and **4***b* during braking and the bottom portion **16**, it is possible to sufficiently secure the rigidity of the inner diameter side cylindrical portion **20** in which the rigidity is particularly required.

[0125] Since a part of the heat transferred from the pads **4***a* and **4***b* to the pad contact portion **24** is also transferred to the columnar ribs **17**, an amount of the heat transferred to the bottom portion **16** is less than that in a case in which no columnar rib **17** is provided. Therefore, the temperature rise of the brake oil between the bottom portion **16** and the cylinder **9** can be prevented.

[0126] In the piston **3** in the present embodiment, in the non-braking state, the portion, which protrudes in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19**, of the inner diameter side cylindrical portion **20** is exposed from the cylinder **9**. Further, the outer peripheral surface of the inner diameter side cylindrical portion **20** is the concave curved surface in which the outer diameter smoothly decreases toward the intermediate portion in the axial direction. Therefore, traveling wind during traveling of the vehicle hits a portion, which is exposed from the cylinder **9**, of the outer peripheral surface of the inner diameter side cylindrical portion **20**, and is guided to the one side in the axial direction (deep side) of the annular gap **21** along the outer peripheral surface of the inner diameter side cylindrical portion **20**. Accordingly, since the outer diameter side cylindrical portion **19** is cooled, the temperature rise of the piston seal **12** is prevented.

[0127] Further, since the outer diameter side cylindrical portion **19** has the tapered surface **23** at the end portion on the other side in the axial direction of the inner peripheral surface, the traveling wind easily enters the annular gap **21**. Therefore, from this point of view as well, the temperature rise of the piston seal **12** is prevented.

[0128] The piston **3** in the present embodiment has the communication holes **28** in the inner diameter side cylindrical portion **20**. Therefore, the traveling wind is guided to the piston internal space **25** through the communication holes **28**. Therefore, the bottom portion **16** can be cooled by using the traveling wind. Therefore, from this point of view as well, the temperature rise of the brake oil can be prevented.

[0129] In the present embodiment, since the communication hole **28** is formed in the portion, in which the phase in the circumferential direction coincides with that of the respective columnar rib **17**, of the inner diameter side cylindrical portion **20**, the traveling wind passing through the communication holes **28** hits the radially outer surfaces of the columnar ribs **17**. Therefore, the columnar ribs **17** are cooled, and the traveling wind is guided to the bottom portion **16** along the columnar ribs **17**. Therefore, the bottom portion **16** is effectively cooled.

[0130] At least a portion of the communication hole **28** is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion **19** and is exposed from the cylinder **9** in the non-braking state. Therefore, an amount of the wind guided to the bottom portion **16** through the communication holes **28** increases. Further, since the communication holes **28** are formed in the inner diameter side cylindrical portion **20**, as indicated by arrows in FIG. **3**, a cooling air introduced into the piston internal space **25** from the outside of the piston **3** through the communication holes **28** can be discharged to the outside of the piston **3** through the communication holes **28**. That is, since the cooling air can be circulated by using the communication holes **28**, heat stored in the piston **3** can be effectively dissipated. Further, as indicated by dashed arrows in FIG. **3**, the cooling air introduced into the piston internal space **25** also passes between the columnar ribs **17**.

[0131] Since the central axis O.sub.28 of the communication hole **28** is inclined with respect to the piston central axis O.sub.3 in the direction approaching the one side in the axial direction when extending toward the radially inner side, and the extension line of the central axis O.sub.28 of the communication hole **28** passes through the side surface on the other side in the axial direction of the bottom portion **16**, the traveling wind passing through the communication holes **28** is efficiently guided to the bottom portion **16**.

[0132] Further, the communication hole **28** is formed in the vicinity of the connection portion, to which the end portion on the other side in the length direction of the columnar rib **17** is connected, of the inner diameter side cylindrical portion **20**, that is, in a portion, which has a sufficiently high strength by being connected to the end portion on the other side in the length direction of the columnar rib **17**, of the inner diameter side cylindrical portion **20**. Therefore, a reduction in the strength of the inner diameter side cylindrical portion **20** caused by forming the communication

holes **28** is efficiently prevented.
[0133] In the piston **3** in the present embodiment, since the heat shielding plate **29** is provided on the radially inner side of the inner diameter side cylindrical portion **20**, the amount of the heat transferred from the pads **4***a* and **4***b* to the bottom portion **16** can be reduced.
[0134] In addition, since the heat shielding plate **29** cannot fall off into the piston internal space **25** by engaging the engagement protrusions **27** with the respective engagement holes **32** formed in the plurality of engagement pieces **31**, it is possible to reduce a cost of the heat shielding plate **29** and to improve the workability of an attachment operation of the heat shielding plate **29**.
Second Embodiment
[0135] A second embodiment of the present disclosure will be described with reference to FIG. **9**.
[0136] In the present embodiment, only a structure of a piston **3***a* is different from the structure of the piston **3** in the first embodiment.
[0137] In the piston **3***a* in the present embodiment, an outer diameter side cylindrical portion **19***a* and an inner diameter side cylindrical portion **20***a*, which form a cylindrical wall portion **15***a*, are formed separately, and the inner diameter side cylindrical portion **20***a* and a plurality of columnar ribs **17***a* are integrally formed.
[0138] In the present embodiment, a piston body **18***a* including the cylindrical wall portion **15***a*, the bottom portion **16***a*, and the plurality of columnar ribs **17***a* is not one component but includes two components. Specifically, the piston body **18***a* includes an outer member **35** and an inner member **36**.
[0139] The outer member **35** has a cup shape. The outer member **35** includes the outer diameter side cylindrical portion **19***a*, and an outer bottom portion **37***a* having a disc-shape, which closes an end portion opening on the one side in the axial direction of the outer diameter side cylindrical portion **19***a*.
[0140] The inner member **36** includes the inner diameter side cylindrical portion **20***a*, an inner bottom portion **37***b* having a disc-shape, which closes an end portion opening on the one side in the axial direction of the inner diameter side cylindrical portion **20***a*, and the plurality of columnar ribs **17***a*.
[0141] The piston body **18***a* in the present embodiment is implemented by fitting the inner member **36** into the outer member **35** by press-fitting. Specifically, the piston body **18***a* is implemented by press-fitting an end portion on the one side in the axial direction of the inner diameter side cylindrical portion **20***a* forming the inner member **36** into an end portion on the one side in the axial direction of the outer diameter side cylindrical portion **19***a* forming the outer member **35** and by overlapping a side surface on the one side in the axial direction of the inner bottom portion **37***b* forming the inner member **36** with no gap on a side surface on the other side in the axial direction of the outer bottom portion **37***a* forming the outer member **35**. Therefore, the bottom portion **16***a* includes the outer bottom portion **37***a* and the inner bottom portion **37***b*, and the annular gap **21** is formed between an inner peripheral surface of the outer diameter side cylindrical portion **19***a* and an inner peripheral surface of the inner diameter side cylindrical portion **20***a* when the inner member **36** is fitted into the outer member **35**.
[0142] In the present embodiment, a thickness t of the columnar rib **17***a* in the radial direction is substantially constant over an entire length. The thickness of the columnar rib **17***a* in the radial direction is larger than radial thicknesses of the outer diameter side cylindrical portion **19***a* and the inner diameter side cylindrical portion **20***a*.
[0143] In the present embodiment, the piston body **18***a* is not one component, but includes two components, the outer member **35** and the inner member **36**. Therefore, the piston body **18***a* can be manufactured by separately manufacturing the outer member **35** and the inner member **36** and then combining the outer member **35** and the inner member **36**, which improves the flexibility of the manufacturing method. Specifically, the outer member **35** and the inner member **36** can be manufactured by machining, injection molding, or the like. Therefore, a manufacturing time can be

shortened, and a manufacturing cost can be reduced.
[0144] Other configurations, operations, and effects are the same as those of the first embodiment.
[0145] As described above, a piston for a disc brake device according to an aspect of the present disclosure is fitted to a cylinder of a caliper forming a disc brake device such as an opposed-piston-type disc brake device or a floating-type disc brake device.
[0146] A piston for a disc brake device according to an aspect of the present disclosure includes: a cylindrical wall portion having a bottomed cylindrical shape as a whole and having a substantially cylindrical shape; a bottom portion configured to close an end portion on one side in an axial direction, which is configured to be disposed on a deep side of the cylinder, of the cylindrical wall portion; and a plurality of columnar ribs disposed on a radially inner side of the cylindrical wall portion.
[0147] The cylindrical wall portion includes an outer diameter side cylindrical portion including a seal sliding contact portion on an outer peripheral surface, an inner diameter side cylindrical portion that is disposed on a radially inner side of the outer diameter side cylindrical portion and that includes a pad contact portion at an end portion on the other side in the axial direction protruding in the axial direction with respect to an end portion on the other side in the axial direction of the outer diameter side cylindrical portion, and an annular gap formed between an inner peripheral surface of the outer diameter side cylindrical portion and an outer peripheral surface of the inner diameter side cylindrical portion.
[0148] The plurality of columnar ribs are disposed to be inclined with respect to a piston central axis, and each have an end portion on one side in a length direction connected to a side surface on the other side in the axial direction of the bottom portion, and an end portion on the other side in the length direction connected to an inner peripheral surface of the inner diameter side cylindrical portion.
[0149] In the piston for the disc brake device according an aspect of the present disclosure, each of the plurality of columnar ribs has the end portion on the other side in the length direction connected to an end portion on the other side in the axial direction of the inner peripheral surface of the inner diameter side cylindrical portion.
[0150] In the piston for the disc brake device according to an aspect of the present disclosure, the plurality of columnar ribs are disposed radially around the piston central axis.
[0151] In the piston for the disc brake device according to an aspect of the present disclosure, each of the plurality of columnar ribs has the end portion on the one side in the length direction connected to an intermediate portion in the radial direction of the side surface on the other side in the axial direction of the bottom portion.
[0152] In the piston for the disc brake device according to an aspect of the present disclosure, the plurality of columnar ribs are disposed at equal intervals in a circumferential direction.
[0153] In the piston for the disc brake device according to an aspect of the present disclosure, the outer diameter side cylindrical portion, the inner diameter side cylindrical portion, and the plurality of columnar ribs are integrally formed.
[0154] In the piston for the disc brake device according to an aspect of the present disclosure, the outer diameter side cylindrical portion and the inner diameter side cylindrical portion are formed separately, and the inner diameter side cylindrical portion and the plurality of columnar ribs are integrally formed.
[0155] In the piston for the disc brake device according to an aspect of the present disclosure, the inner diameter side cylindrical portion has a communication hole penetrating in the radial direction.
[0156] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a portion, in which a phase in the circumferential direction coincides with that of the columnar rib, of the inner diameter side cylindrical portion.
[0157] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a vicinity of a connection portion, to which the end portion on the

other side in the length direction of the columnar rib is connected, of the inner diameter side cylindrical portion.
[0158] In the piston for the disc brake device according to an aspect of the present disclosure, the communication hole is formed in a portion adjacent to one side in the axial direction of the connection portion.
[0159] In the piston for the disc brake device according to an aspect of the present disclosure, at least a portion of the communication hole is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion.
[0160] In the piston for the disc brake device according to an aspect of the present disclosure, a central axis of the communication hole is inclined with respect to the piston central axis in a direction approaching the one side in the axial direction when extending toward the radially inner side.
[0161] In the piston for the disc brake device according to an aspect of the present disclosure, a heat shielding plate is provided on a radially inner side of the inner diameter side cylindrical portion, and the heat shielding plate covers the side surface on the other side in the axial direction of the bottom portion in a state of being in contact with the plurality of columnar ribs.
[0162] In the piston for the disc brake device according to an aspect of the present disclosure, the heat shielding plate is configured to be engaged with the plurality of columnar ribs.
[0163] In the piston for the disc brake device according to an aspect of the present disclosure, each of the plurality of columnar ribs has an engagement protrusion on a portion in the length direction thereof, the heat shielding plate has engagement holes, and the engagement protrusions are configured to be engaged with the respective engagement holes.
[0164] A disc brake device according to an aspect of the present disclosure includes: a caliper having a cylinder; and a piston fitted into the cylinder.
[0165] In the disk brake device according to an aspect of the present disclosure, the piston is the piston for the disc brake device according to an aspect of the present disclosure.
[0166] According to the piston for a disc brake device according to an aspect of the present disclosure, it is possible to prevent a temperature rise of a brake oil, to make it difficult for heat to be transmitted to a piston seal, to reduce weight, and to ensure the rigidity.

Claims

1. A piston for a disc brake device, which is configured to be fitted into a cylinder of a caliper forming the disc brake device, the piston for the disc brake device comprising: a cylindrical wall portion having a bottomed cylindrical shape as a whole and having a substantially cylindrical shape; a bottom portion configured to close an end portion on one side in an axial direction, which is configured to be disposed on a deep side of the cylinder, of the cylindrical wall portion; and a plurality of columnar ribs disposed on a radially inner side of the cylindrical wall portion, wherein the cylindrical wall portion includes an outer diameter side cylindrical portion including a seal sliding contact portion on an outer peripheral surface, an inner diameter side cylindrical portion that is disposed on a radially inner side of the outer diameter side cylindrical portion and that includes a pad contact portion at an end portion on the other side in the axial direction protruding in the axial direction with respect to an end portion on the other side in the axial direction of the outer diameter side cylindrical portion, and an annular gap formed between an inner peripheral surface of the outer diameter side cylindrical portion and an outer peripheral surface of the inner diameter side cylindrical portion, and the plurality of columnar ribs are disposed to be inclined with respect to a piston central axis, and each have an end portion on one side in a length direction connected to a side surface on the other side in the axial direction of the bottom portion, and an end portion on the other side in the length direction connected to an inner peripheral surface of the inner diameter side

cylindrical portion.
2. The piston for the disc brake device according to claim 1, wherein each of the plurality of columnar ribs has the end portion on the other side in the length direction connected to an end portion on the other side in the axial direction of the inner peripheral surface of the inner diameter side cylindrical portion.
3. The piston for the disc brake device according to claim 1, wherein the plurality of columnar ribs are disposed radially around the piston central axis.
4. The piston for the disc brake device according to claim 1, wherein each of the plurality of columnar ribs has the end portion on the one side in the length direction connected to an intermediate portion in the radial direction of the side surface on the other side in the axial direction of the bottom portion.
5. The piston for the disc brake device according to claim 1, wherein the plurality of columnar ribs are disposed at equal intervals in a circumferential direction.
6. The piston for the disc brake device according to claim 1, wherein the outer diameter side cylindrical portion, the inner diameter side cylindrical portion, and the plurality of columnar ribs are integrally formed.
7. The piston for the disc brake device according to claim 1, wherein the outer diameter side cylindrical portion and the inner diameter side cylindrical portion are formed separately, and the inner diameter side cylindrical portion and the plurality of columnar ribs are integrally formed.
8. The piston for the disc brake device according to claim 1, wherein the inner diameter side cylindrical portion has a communication hole penetrating in the radial direction.
9. The piston for the disc brake device according to claim 8, wherein the communication hole is formed in a portion, in which a phase in the circumferential direction coincides with that of the columnar rib, of the inner diameter side cylindrical portion.
10. The piston for the disc brake device according to claim 8, wherein the communication hole is formed in a vicinity of a connection portion, to which the end portion on the other side in the length direction of the columnar rib is connected, of the inner diameter side cylindrical portion.
11. The piston for the disc brake device according to claim 10, wherein the communication hole is formed in a portion adjacent to one side in the axial direction of the connection portion.
12. The piston for the disc brake device according to claim 8, wherein at least a portion of the communication hole is located on the other side in the axial direction with respect to the end portion on the other side in the axial direction of the outer diameter side cylindrical portion.
13. The piston for the disc brake device according to claim 8, wherein a central axis of the communication hole is inclined with respect to the piston central axis in a direction approaching the one side in the axial direction when extending toward the radially inner side.
14. The piston for the disc brake device according to claim 1, wherein a heat shielding plate is provided on a radially inner side of the inner diameter side cylindrical portion, and the heat shielding plate covers the side surface on the other side in the axial direction of the bottom portion in a state of being in contact with the plurality of columnar ribs.
15. The piston for the disc brake device according to claim 14, wherein the heat shielding plate is configured to be engaged with the plurality of columnar ribs.
16. The piston for the disc brake device according to claim 15, wherein each of the plurality of columnar ribs has an engagement protrusion on a portion in the length direction thereof, the heat shielding plate has engagement holes, and the engagement protrusions are configured to be engaged with the respective engagement holes.
17. A disc brake device comprising: a caliper having a cylinder; and a piston fitted into the cylinder, wherein the piston is the piston for the disc brake device according to claim 1.